Virtual Weapons in Real Life

Video games have always had over-the-top weapons. Is there anything similar in the real world?

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If you're reading this magazine, you've probably played video games at least once in your life, be it on those old Sega controllers or on today's mean consoles. Over the years, games have evolved from the simple but highly addictive *Dave* to *Contra* and on to today's near-realistic-physics-and-animation games and massively multi-player online (MMO) games like *Gears Of War*. As these games have developed, so have the weapons: from the simple "gun" in *Dave*, we've evolved to the "Big F***ing Gun" in the *Quake* series, and the "Gravity Gun" in *Half-Life 2*. We thought we'd dig into classified data to see if any real-world weapons in development today are similar to those in video games. Could these have been... inspired by the games? In any case, without further ado, our expert findings:

A Command & Conquer soldier in full armour

Second Skin

Homo sapiens isn't the swiftest creature on Earth. Most of us are limited by how much weight we can lug around.

In games such as *S.T.A.L.K.E.R.* and *Half-Life 2*, the powered exoskeleton is an external skin-like framework attached to a solider. It incorporates a unit that supplies energy for physical movement and other functions that the exoskeleton performs. In the *Metroid* series of video games, released in stone-age 1986, Samus Aran, the lead character, wears an exoskeleton enhanced with a cannon attached to the arm. The suit gives the player special abilities like rolling into a ball or performing very high spinning jumps. And yes, it also allows the player to stay underwater or in the vacuum of space almost indefinitely!

With this thing around, joining the armed forces would be so much cooler...

And now, the United States Defense Advanced Research Projects Agency (DARPA) is investing $50 million to develop an exoskeletal suit for ground troops. Imagine turning ordinary soldiers into super-troops who can leap tall objects in a single bound and run at high speeds (unsaid reference to Superman purely coincidental). The details of this program are "blacked out" and classified, and information about these wearable machines is hard to come by. However, DARPA has set some expectations for these exoskeletal machines. Here's what they are expected to do for soldiers:

Increased strength: This one is a no-brainer... We're excited about the prospect of soldiers being able to remove large obstacles from their paths without using heavy machinery. In the future, the exoskeletons will include heavier body armour and protection from ballistic explosions from grenades and rocket launchers too. Coincidentally, in the 1960s, GE in collaboration with the US military jointly developed an exoskeleton named "Hardiman," which made lifting 300 pounds feel like lifting 15 pounds!

Increased speed: Akin to making our bots run around like rabbits on an unknown map, soldiers are often expected to carry up to 150 pounds of supplies in their backpacks over long distances. Even the most hardened of troops can't get very far carrying that much on their backs... It's not yet clear as to how fast DARPA's exoskeleton will be able traverse over different terrain, but an independently-developed body amplifier, christened the "SpringWalker," has been tested to move over terrain at speeds faster than 16 kmph (the average walking speed is 3.5kmph)!

Leap great heights and distances: Imagine a marching legion of troops wearing these exoskeletons, jumping over a wide crevice in a mountain. and moving forward relentlessly! It's yet unclear as to just how far or high soldiers will be able to jump, but officials would like the exoskeleton to give soldiers these Superman-like abilities.

Just one example of what the future solider might look like

Communications integration: The exoskeletal machines will also feature sensors and GPS receivers. In the future, every soldier on the battlefield will have mission-centric information like the terrain they are crossing and how to navigate their way to specific locations in the war zone right inside their helmets. DARPA is also rumoured to be developing fabrics that could be used with the exoskeletons to monitor heart and breathing rates so any health-related issues can be detected on time. Of course, instant medkits haven't been developed yet...